Ex No: 15

Endpoint security monitoring

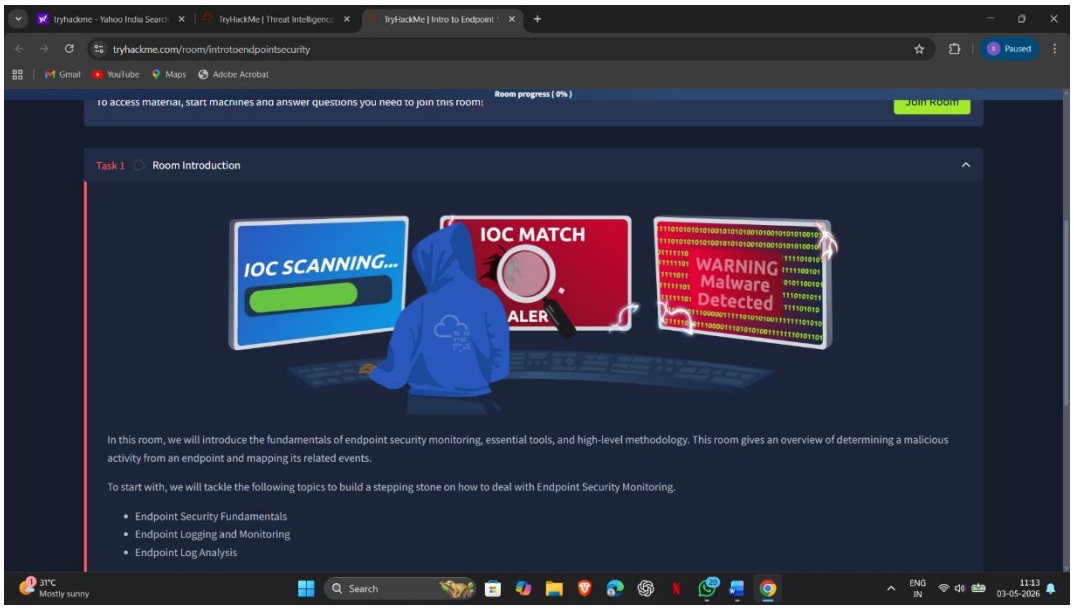
Aim:

To monitor and analyze endpoint activities to detect, prevent, and respond to security threats using endpoint security tools in a SOC environment.

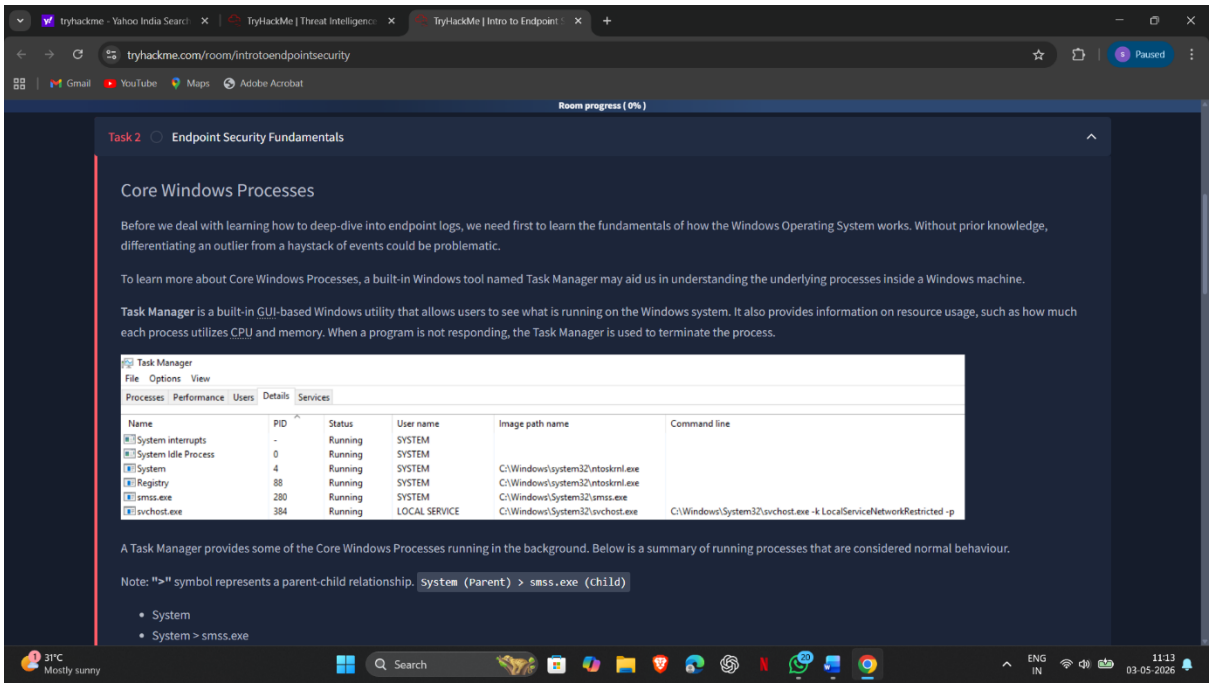
Procedure:

1. Open a web browser and go to TryHackMe, then log in with your account.
2. Search for the room related to Endpoint Security Monitoring and open it.
3. Read the introduction to understand endpoint security concepts such as antivirus and Endpoint Detection and Response (EDR).
4. Start the room by clicking "Start" or launch the machine if required.
5. Study how endpoint security tools monitor processes, file activities, and system behavior.
6. Analyze alerts and logs generated by endpoint security solutions to identify suspicious or malicious activities.
7. Investigate Indicators of Compromise (IoCs) such as unusual processes, file changes, or unauthorized access.
8. Perform the given tasks and answer all questions based on your analysis.
9. Document important observations, tools used, and capture screenshots if required for record submission.
10. Complete all tasks until the room shows 100% completion.
11. Log out from TryHackMe after finishing the experiment.

Task 1: Introduction



Task 2: Endpoint Security Fundamentals



Task 2 Endpoint Security Fundamentals

Core Windows Processes

Before we deal with learning how to deep-dive into endpoint logs, we need first to learn the fundamentals of how the Windows Operating System works. Without prior knowledge, differentiating an outlier from a haystack of events could be problematic.

To learn more about Core Windows Processes, a built-in Windows tool named Task Manager may aid us in understanding the underlying processes inside a Windows machine.

Task Manager is a built-in GUI-based Windows utility that allows users to see what is running on the Windows system. It also provides information on resource usage, such as how much each process utilizes CPU and memory. When a program is not responding, the Task Manager is used to terminate the process.

Task Manager

File Options View

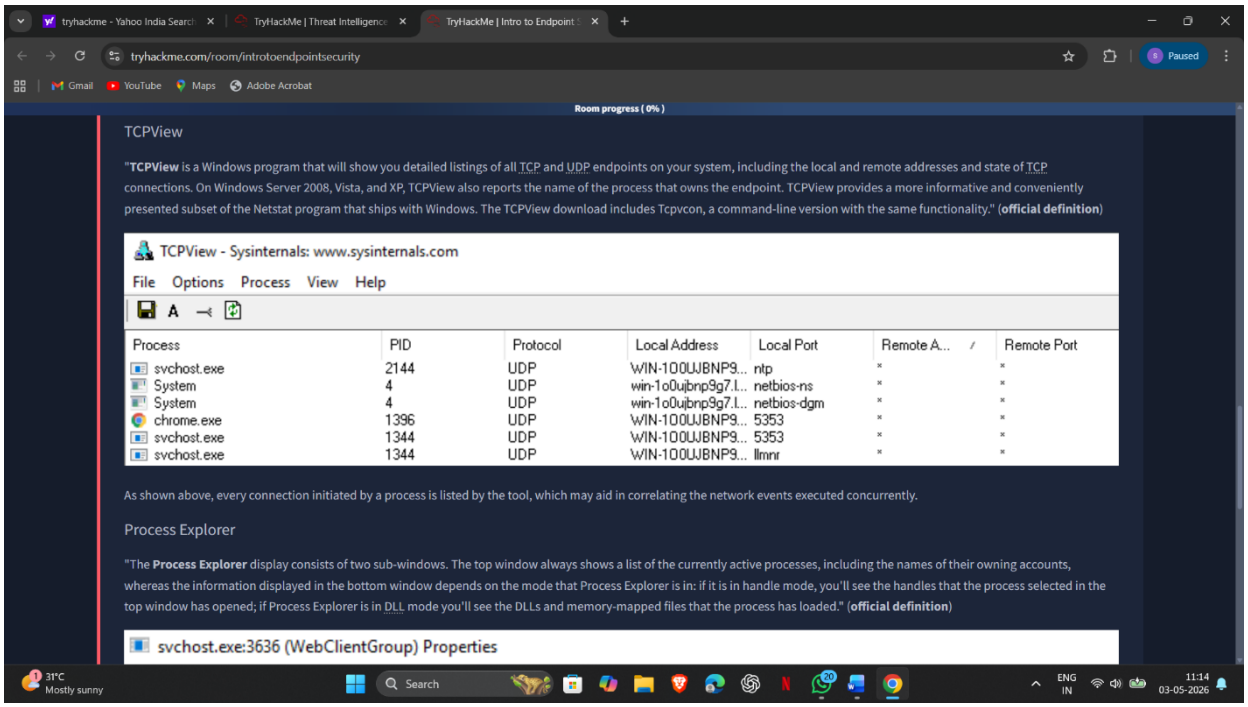
Processes Performance Users Details Services

Name	PID	Status	User name	Image path name	Command line
System interrupts	-	Running	SYSTEM		
System Idle Process	0	Running	SYSTEM		
System	4	Running	SYSTEM	C:\Windows\system32\ntoskrnl.exe	
Registry	88	Running	SYSTEM	C:\Windows\system32\ntoskrnl.exe	
smss.exe	280	Running	SYSTEM	C:\Windows\System32\smss.exe	
svchost.exe	384	Running	LOCAL SERVICE	C:\Windows\System32\svchost.exe	C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p

A Task Manager provides some of the Core Windows Processes running in the background. Below is a summary of running processes that are considered normal behaviour.

Note: ">" symbol represents a parent-child relationship. `System (Parent) > smss.exe (Child)`

- System
- System > smss.exe



TCPView

"**TCPView** is a Windows program that will show you detailed listings of all TCP and UDP endpoints on your system, including the local and remote addresses and state of TCP connections. On Windows Server 2008, Vista, and XP, TCPView also reports the name of the process that owns the endpoint. TCPView provides a more informative and conveniently presented subset of the Netstat program that ships with Windows. The TCPView download includes Tcpvcon, a command-line version with the same functionality." (**official definition**)

TCPView - Sysinternals: www.sysinternals.com

File Options Process View Help

Process	PID	Protocol	Local Address	Local Port	Remote A...	Remote Port
svchost.exe	2144	UDP	WIN-100UJBNP9...	ntp	*	*
System	4	UDP	win-1o0ujbnp9g7.l...	netbios-ns	*	*
System	4	UDP	win-1o0ujbnp9g7.l...	netbios-dgm	*	*
chrome.exe	1396	UDP	WIN-100UJBNP9...	5353	*	*
svchost.exe	1344	UDP	WIN-100UJBNP9...	5353	*	*
svchost.exe	1344	UDP	WIN-100UJBNP9...	llmnr	*	*

As shown above, every connection initiated by a process is listed by the tool, which may aid in correlating the network events executed concurrently.

Process Explorer

"The **Process Explorer** display consists of two sub-windows. The top window always shows a list of the currently active processes, including the names of their owning accounts, whereas the information displayed in the bottom window depends on the mode that Process Explorer is in: if it is in handle mode, you'll see the handles that the process selected in the top window has opened; if Process Explorer is in DLL mode you'll see the DLLs and memory-mapped files that the process has loaded." (**official definition**)

svchost.exe:3636 (WebClientGroup) Properties

Task 3: Endpoint Logging and Monitoring

Room progress (0%)

Task 3 Endpoint Logging and Monitoring

From the previous task, we have learned basic knowledge about the Windows Operating system in terms of baseline processes and essential tools to analyze events and artefacts running on the machine. However, this only limits us from observing real-time events. With this, we will introduce the importance of endpoint logging, which enables us to audit significant events across different endpoints, collect and aggregate them for searching capabilities, and better automate the detection of anomalies.

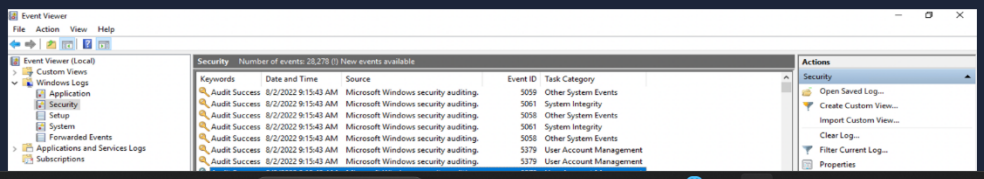
Windows Event Logs

The Windows Event Logs are not text files that can be viewed using a text editor. However, the raw data can be translated into XML using the Windows API. The events in these log files are stored in a proprietary binary format with a .evt or .evtx extension. The log files with the .evtx file extension typically reside in `C:\Windows\System32\winevt\Logs`.

There are three main ways of accessing these event logs within a Windows system:

1. **Event Viewer** (GUI-based application)
2. **Wevtutil.exe** (command-line tool)
3. **Get-WinEvent** (PowerShell cmdlet)

An example image of logs viewed using the **Event Viewer** tool is shown below.



Room progress (0%)

Sysmon

Sysmon, a tool used to monitor and log events on Windows, is commonly used by enterprises as part of their monitoring and logging solutions. As part of the Windows Sysinternals package, Sysmon is similar to Windows Event Logs with further detail and granular control.

Sysmon gathers detailed and high-quality logs as well as event tracing that assists in identifying anomalies in your environment. It is commonly used with a security information and event management (SIEM) system or other log parsing solutions that aggregate, filter, and visualize events.

Lastly, Sysmon includes 27 types of Event IDs, all of which can be used within the required configuration file to specify how the events should be handled and analyzed. An excellent example of a configuration file auditing different Event IDs created by SwiftOnSecurity is linked here.

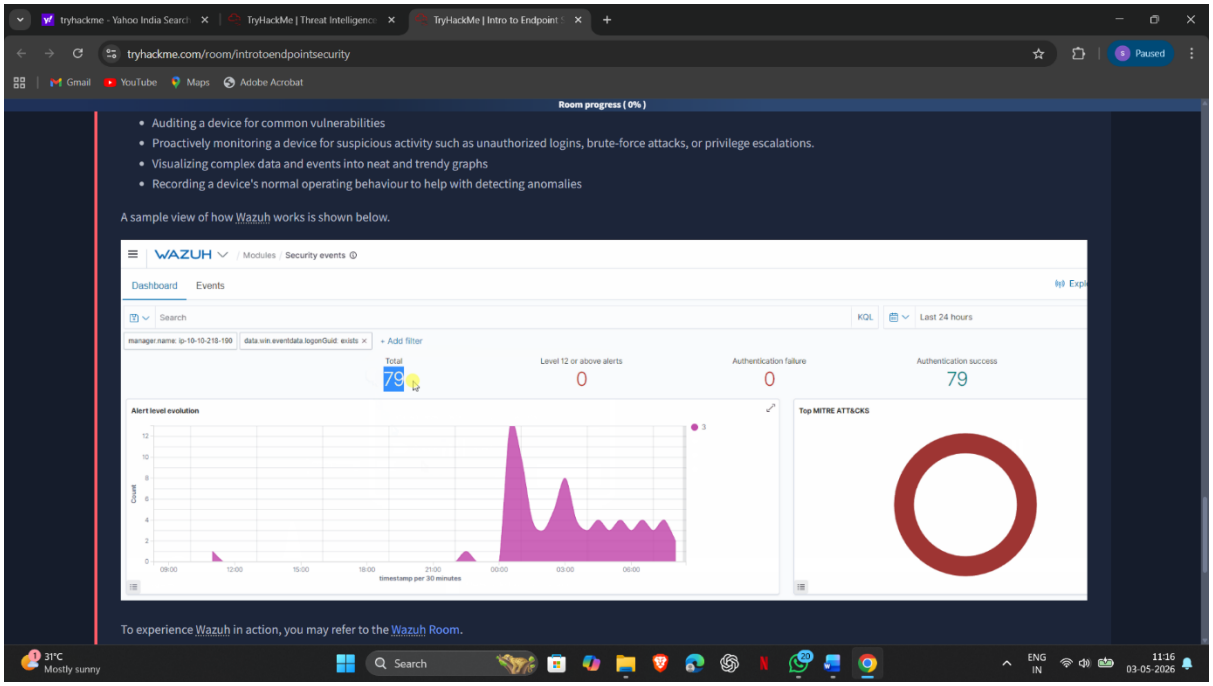
The image below shows a sample set of Sysmon logs viewed using an **Event Viewer**.

Operational Number of events: 222 (!) New events available

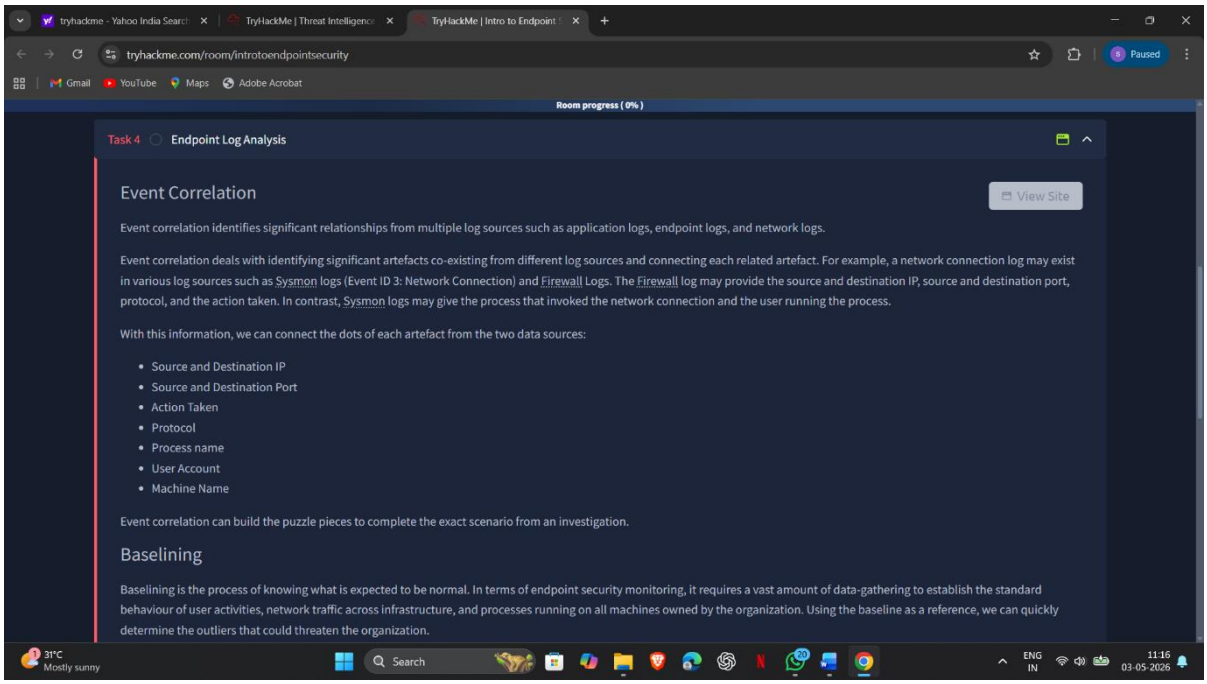
Level	Date and Time	Source	Event ID	Task Category
Information	12/18/2020 1:35:12 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:36:31 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:43:59 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:43:59 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:36:44 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:46:44 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:41:44 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:36:44 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:37:21 AM	Sysmon	11	File created (rule: FileCreate)
Information	12/18/2020 1:41:43 AM	Sysmon	11	File created (rule: FileCreate)

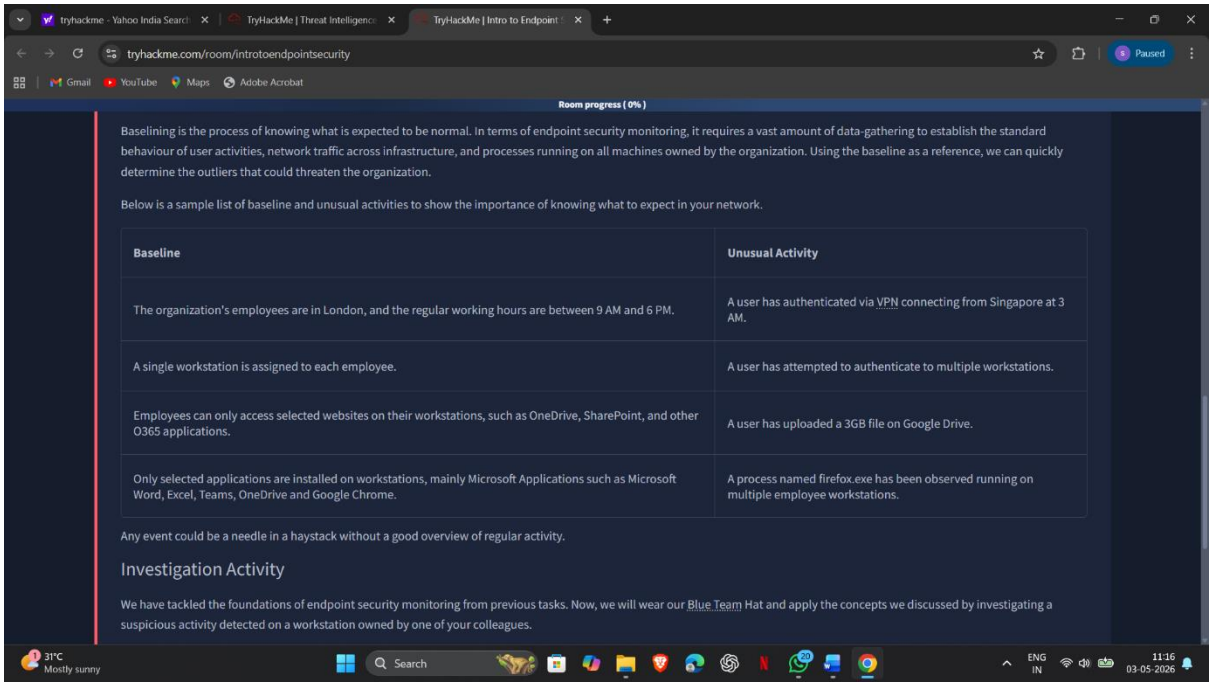
To learn more about Sysmon, you may refer to the Sysmon Room.

OSQuery



Task 4: Endpoint Log analysis





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Baselining is the process of knowing what is expected to be normal. In terms of endpoint security monitoring, it requires a vast amount of data-gathering to establish the standard behaviour of user activities, network traffic across infrastructure, and processes running on all machines owned by the organization. Using the baseline as a reference, we can quickly determine the outliers that could threaten the organization.

Below is a sample list of baseline and unusual activities to show the importance of knowing what to expect in your network.

Baseline	Unusual Activity
The organization's employees are in London, and the regular working hours are between 9 AM and 6 PM.	A user has authenticated via VPN connecting from Singapore at 3 AM.
A single workstation is assigned to each employee.	A user has attempted to authenticate to multiple workstations.
Employees can only access selected websites on their workstations, such as OneDrive, SharePoint, and other O365 applications.	A user has uploaded a 3GB file on Google Drive.
Only selected applications are installed on workstations, mainly Microsoft Applications such as Microsoft Word, Excel, Teams, OneDrive and Google Chrome.	A process named firefox.exe has been observed running on multiple employee workstations.

Any event could be a needle in a haystack without a good overview of regular activity.

Investigation Activity

We have tackled the foundations of endpoint security monitoring from previous tasks. Now, we will wear our Blue Team Hat and apply the concepts we discussed by investigating a suspicious activity detected on a workstation owned by one of your colleagues.

Result:

Successfully monitored and analyzed endpoint activities to detect potential threats. Gained practical knowledge of identifying malicious behavior, interpreting endpoint security alerts, and responding to incidents using endpoint security tools in a SOC environment.